

**Third Semester B. Tech. (Electronics and Communication
Engineering) Examination**

APPLIED MATHEMATICS–III

Time : 3 Hours]

[Max. Marks : 50

Instructions to Candidates :—

- (1) All questions carry equal marks.
- (2) Use of Non – programmable calculator is permitted.

1. Solve any **Two** :

(a) Find the Laplace Transform of :

(i) $e^{-3t}(2 \cos 5t - 3 \sin 5t)$

(ii) $\frac{\cos t}{t}$

5(CO1)

(b) Find the Inverse Laplace transform of $\log \frac{s+1}{s-1}$.

5(CO1)

(c) Use the Transform method to solve $\frac{d^2x}{dt^2} - 2 \frac{dx}{dt} + x = e^t$ with $x = 2$,

$$\frac{dx}{dt} = -1 \text{ at } t = 0$$

5(CO1)

2. Solve any **Two** :(a) Find z-Transform of $\sin (3k + 5)$, $k \geq 0$.

5(CO2)

(b) Find z-Transform inverse of $\frac{z}{(z+2)(z+3)}$.

5(CO2)

(c) Using the z-Transform, solve $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$ with $u_0 = 0$, $u_1 = 1$.

5(CO2)

3. Solve :

(a) Find the Fourier sine transform of $e^{-|x|}$, $x > 0$ and hence show that

$$\int_0^{\infty} \frac{x \sin \pi x}{1+x^2} dx = \frac{\pi}{2} e^{-\pi}. \quad 5(\text{CO3})$$

(b) Using the Fourier integral, show that

$$\int_0^{\infty} \frac{1 - \cos \pi \lambda}{\lambda} \sin \lambda x d\lambda = \begin{cases} \frac{\pi}{2}, & 0 < x < \pi \\ 0, & x > \pi \end{cases} \quad 5(\text{CO3})$$

4. Solve any **Two** :

(a) Solve the following Lagrange's linear equation $xq = yp + xe^{x^2+y^2}$

$$\text{where, } p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y} \quad 5(\text{CO5})$$

(b) Solve $\frac{\partial^3 z}{\partial x^3} - \frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x} + 3x^2y$ 5(CO5)

(c) Solve by using the method of separation of variable, $4 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$
given that $u = 3e^{-y} - e^{-5y}$ where, $x = 0$. 5(CO5)

5. Solve any **Two** :

(a) Test whether the following function is analytic or not.

$$f(z) = r^n \cos n\theta + i r^n \sin n\theta \quad 5(\text{CO4})$$

(b) Find the Laurent series expansion of the function $f(z) = \frac{1}{(z-1)(z-2)}$
in the region $1 < |z| < 2$ 5(CO4)

(c) Evaluate $\oint_C \frac{e^z}{\cosh z} dz$, where C is a circle $|z| = 5$ 5(CO4)

